

Optimal Settings for Multiple Groups of Smart Inverters on Secondary Systems Using Autonomous Control

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Abstract — To support distribution planning and operations engineers, this paper describes an optimal methodology for specifying smart inverter settings that will improve PV integration into distribution systems. The dynamic nature of the distribution system requires that a range of load and solar conditions be considered for this purpose. Time-series analysis was performed using OpenDSS to determine the overall impact on the distribution feeder. For the feeder investigated, the use of smart inverters was evaluated for mitigating potential voltage-related issues by reducing the magnitude and/or number of ANSI voltage violations as well as meeting other objectives like increasing in substation power factor and reducing distribution system losses.

Index Terms — smart inverters, secondary systems, autonomous, solar PV

I. INTRODUCTION

The roll out of solar PV to residential customers within the distribution territory of most utilities is increasing daily. With this increase, comes technical and regulatory issues. The Electric Power Research Institute (EPRI) has done extensive work on evaluating smart inverter impacts and determining appropriate settings to inform decisions for policy making, distribution planning and operations, to name a few. Nowadays, inverter manufacturers are compelled to enhance the functionalities of various inverters irrespective of their size and configuration. This gives distribution engineers considerable configuration flexibility in how the inverters respond and interact with the grid. To improve grid integration, the distribution engineer can choose from default, “out-of-the-box” type settings or choose to optimize the configuration settings based upon location-specific grid characteristics. Regardless, the objective is to improve the integration of Distributed Energy Resources (DER) by reducing or eliminating adverse impacts caused by the DER and in some cases improve overall distribution performance. Methods have been proposed in [1,2,3,4] to determine site specific inverter settings but results show how those settings are highly dependent on the specific scenarios analyzed.

The work herein summarized mainly focuses on smart inverters connected directly on secondary systems. The use case explained in this paper will describe the technique to determine optimal smart inverter settings where several smart

inverter capabilities can be deployed at residential customers’ sites.

This paper describes methods to determine optimal settings for effective use of advanced inverters using the several inverter functions. The end goal is to recommend settings for the advanced inverter functions to achieve specified objectives. The layout of the paper is to initially describe the use case model, explain the advanced inverter functionalities and then highlight the methodology to determine the inverter settings; lastly the paper will explain the results of the settings derived based on the set objectives.

II. BACKGROUND

A. Use Case Feeder model

The original feeder model was developed from Geographic Information System (GIS) data provided by the utility and transcribed into the Open-source Distribution System Simulator modeling environment (OpenDSS).

The precision of the model used in the simulation has a direct impact on how accurate the simulated results will be to the field response when the inverter control settings are effected in the field [5]. Hence, much care is given to modeling and validation of the feeder. The characteristics of the feeder discussed are summarized in Table I.

TABLE I. STUDY FEEDER CHARACTERISTICS.

Feeder	Nominal Voltage (kV)	Peak Load (kW)	Load power factor	Capacitors (kvar)
Feeder XYZ	12.47	9900	0.9	7200

This developed model is more extensive than those typically used for planning, including distribution transformers, secondary service drops, and individual customer loads. Customer meter (both PV and billing) recordings from 2015 were used to create the time-series load profiles used in the model. The net load profile was compared against Energy Management System (EMS) records as verification. The voltage profile shows the snap shot of the peak load of the feeder. For verification, the normal operating voltages (or voltage profile) and expected fault currents calculated by the

model at various points along the feeder were compared against planning values obtained from the utility.

The dynamic nature of the distribution system requires that a range of load and solar conditions be evaluated. This steady-state evaluation was also accompanied by a “time-series” validation. The time series validation includes populating the load shape of all the solar PVs for one calendar year. It further adds a representation of regulators, capacitor banks, and their associated control systems into the model. This goal is to effectively mimic how the devices are operated in practice, either by automated control or operator [5]. The primary voltages per phase are represented in Fig 1.

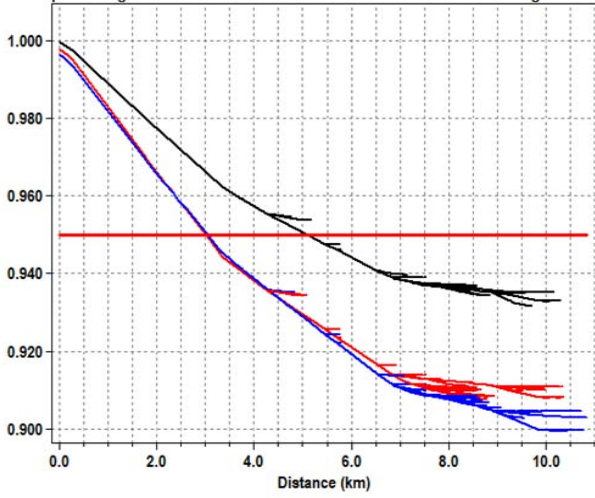


Fig. 1. Feeder L-N voltage profile in OpenDSS.

Voltage control on the feeder is achieved through the use of switched capacitors on the feeder. There is no feeder head voltage regulator and the voltage set point at the secondary bus of the 69/12.47, 25MVA transformer is 1.01pu, which is fixed rather than regulated based upon loading. The transformer impedance is 8.98%. The residential loads have a power factor of 0.9 (lagging) while all commercial loads have a 0.85 power factor (lagging). In terms of PV generation to load composition, the feeder peak PV and most back fed PVs occurred in April, 2015, while the heavy load on the feeder occurred in May 2015. The solar PV and load composition of the feeder is described in Table II.

TABLE II. PEAK LOAD AND SOLAR PV INFORMATION ON FEEDER XYZ

Feeder name	Loading condition	Abbreviation	kW
Feeder XYZ	Peak (PV-PV1, Net load-NL1, Total load-TL1) kW	PV1	2700
		NL1	-300
		TL1	2400
	Most back fed (PV-PV2, Net load-NL2, Total load-TL2) kW	PV2	2650
		NL2	-500
		TL2	2100
	Heavy load (PV-PV3, Net load-NL3, Total load-TL3) kW	PV3	770
		NL3	9100
		TL3	9900

It is evident that the total number of existing inverters on this feeder is large enough to cause reverse power flow into the transmission system on the most backfed day. In addition to the existing inverters, the utility customers were approached to participate in voluntary smart inverter roll out to check the impact on the feeder performance on existing distribution system. After the program, the percentage of the total residential smart inverters (in kVA) size to the existing feeder peak load is less than 5%. Each residential smart inverter is rated at 6kVA.

B. Smart inverter functionalities

There are several smart inverter functionalities that can be simulated and analyzed on any distribution feeder. For this particular analysis, the functions considered include inductive power factor control, volt-var control, and volt-watt control [1,4,6]. These functions can be summarized to mean the following:

Power Factor – Inductive output based on the real power generated by the PV: allows the power factor of Distributed Energy Resource (DER) to be set at a fixed value.

Volt-Var – Inductive or Capacitive output based on the voltage at the PV inverter: allows the DER to manage its own reactive power output in response to local service voltage.

Volt-Watt – Maximum real power output based on the voltage at the PV inverter: allows the DER to manage power output based on voltage.

In addition, a range of possibilities has been chosen to represent a wide spectrum of possibilities. An example set of volt-var control curves are shown in Fig. 2 to Fig. 4. There are three types which include Fig. 2 no dead band, Fig. 3 with dead band, and Fig. 4 inductive mode only.

Although these are three types of curves, other possibilities could include dead bands off the y=0 axis, shifts further left/right on the x-axis, or nonlinear variation in the inductive/capacitive regions.

Each of these functions can have a direct impact on the operation of the distribution system. Additional advanced inverter functions such as Frequency-Watt, Voltage Ride-Through, and Frequency Ride-Through have a more direct impact on the bulk power system [4]. For this reason, they are not considered for this analysis.

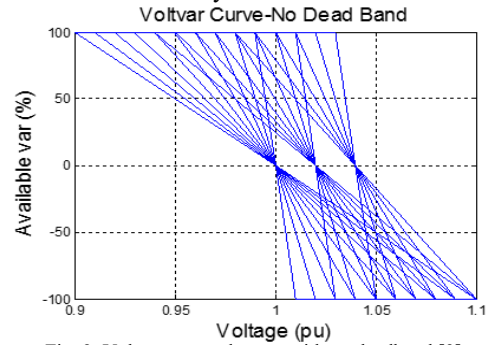


Fig. 2. Volt-var control curve with no deadband [3].

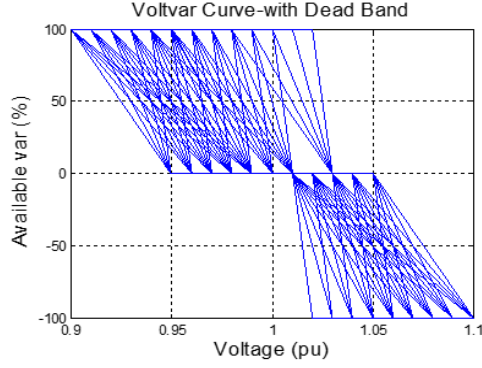


Fig. 3. Volt-var control with deadband [3].

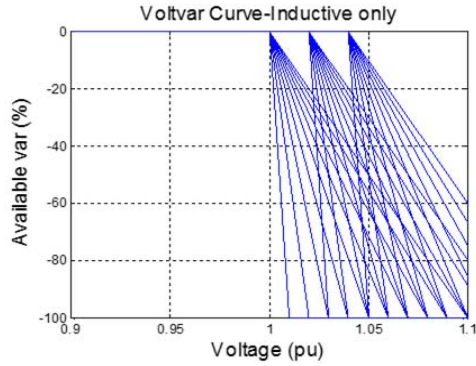


Fig. 4. Volt-var control in Inductive mode only [3].

C. Autonomous vs integrated control

Autonomous inverter control is a situation in which the measurement of the inverter outputs is based on the inverter node voltage. In this project, each individual inverter control is based on the primary node voltage at the interconnect location.

An integrated inverter control is a Distributed Energy Resource Management System (DERMS) like scenario based on the electrical proximity; to group, monitor, measure, operate and coordinate all the smart inverters within the distribution feeder. It must be noted that DERMS option wasn't analyzed in this paper.

D. Load and solar probability of daily variations

This research is leveraging previous investigations which explain the dependence of suitable smart inverter settings on solar characteristics and load levels [1,2,3,4]. Some settings are best suited for the severity of cloud-induced power output swings while others perform best on clear days. The appropriate setting also has a heavy dependence on load level. Therefore, all the possible solar categories based on 365 days were analyzed on actual data for the site where this research was conducted. Two load levels were then overlaid on these solar day categories.

Solar categorization - The clearness index and variability index are used to choose the appropriate solar day categories. These two variables quantify irradiance variability in order to ensure the days chosen for simulation cover the range of possibilities. The variability index is defined as:

$$VI = \frac{\sum_{k=2}^n \sqrt{(GHI_k - GHI_{k-1})^2 + \Delta t^2}}{\sum_{k=2}^n \sqrt{(CSI_k - CSI_{k-1})^2 + \Delta t^2}}$$

where the sequences GHI and CSI are the global horizontal irradiance and clear sky irradiance, respectively, sampled every time Δt [10]. The clearness index is defined as the ratio of solar energy measured on a given surface to the theoretical maximum energy on that same surface during a clear sky day [6,7].

Load categorization - For each of the three solar categories, two load categories are considered based on the average daily load.

III. METHODOLOGY AND ANALYSIS FRAMEWORK

The core framework to determine the smart inverter setting on a secondary system is a very rigorous process. Since the smart inverters have various control settings, it is always advisable to highlight the key and applicable objectives to be measured as the critical evaluation criteria. A summary of the feeder impact criteria as described previously [3] is shown in Table III.

TABLE III. FEEDER IMPACT CRITERIA [3]

Feeder Impact Criteria	Used in Analysis
Minimum Feeder Head Power Factor	
Tap Operations	
Cap Operations	
Feeder Losses (kWh)	X
Feeder Consumption (MWh)	
Max Feeder-Wide Voltage (pu)	
Time Above ANSI Max Voltage (sec)	
Min Feeder-Wide Voltage (pu)	
Time Below ANSI Min Voltage (sec)	X
Difference between Feeder-Wide Max and Min	X
Max Feeder Head Voltage (pu)	
Mean Feeder Head Voltage (pu)	
Min Feeder Head Voltage (pu)	
Max Feeder End Voltage (pu)	
Mean Feeder End Voltage (pu)	
Min Feeder End Voltage (pu)	
Max PCC Voltage (pu)	
Mean PCC Voltage (pu)	
Min PCC Voltage (pu)	
Voltage Variability Index at inverter terminals	X

Each setting influences key system variables in a slightly different way. The modeler has to specify the appropriate weighting factor for several key elements or objectives.

Secondly, a detailed time-series simulation is performed. This involves simulating a circuit during sequential time intervals, and carrying the results of one simulation over to the next. This allows controller interaction to be tracked over the course of the simulation. For investigating smart inverter functions, simulations were run every minute (or 1440 total per day) for each of six sample days. The six combinations are off-peak clear day, peak clear day, off peak variable day, peak variable day, off-peak overcast day and peak over cast day. Each sample consisted of one of two load conditions, and one of three solar profiles (clear, variable, and overcast). Once the six sample days are evaluated, the inverter settings are switched to the next candidate and the process begins again [5].

Lastly, the selection of the best settings is done by ranking the control setting based on the performance metrics or objectives and then down-selecting based on voltage limits.

In all instances, the ranking is weighted based on the feeder impact criteria metric and also the solar/load condition. Table IV shows the weighted average of days considered over one year.

TABLE IV. LOAD SOLAR PROBABILITY WEIGHTINGS

Day	Weighting
Variableday_peak	34.8000
Variableday_offpeak	81.2000
Overcastday_peak	0.6000
Overcast_offpeak	1.4000
Clearday_peak	74.4000
Clearday_offpeak	173.6000

The weighting, based on feeder impact, is applied because multiple feeder impact metrics can be examined simultaneously. The weighing is also adjusted based on solar/load condition to determine the overall impact over a longer time horizon where some conditions are more likely to occur than others [4,8].

Once all of the candidate settings have been evaluated, those that provided the best “score” when evaluated against the criteria above are kept as “optimal” settings.

A summary of the analysis framework is presented in Fig 5.

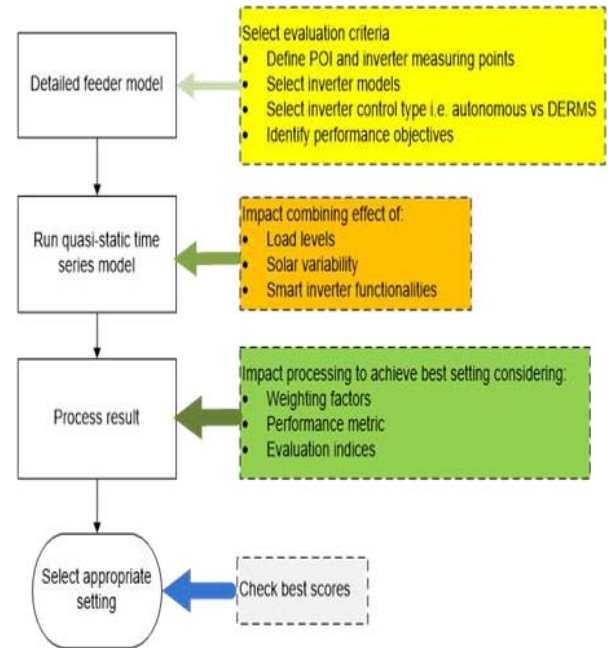


Fig 5: Smart inverter analysis framework

IV. RESULTS

For all the smart inverters connected to the roof tops on the secondary system analyzed, several key performance metrics were evaluated.

These include the following:

- Voltage difference: the difference between the feeder wide maximum and minimum voltages. This determines if the smart inverters can flatten the primary feeder voltage profile.
- Voltage variability across the feeder. In order to confirm the number of variations due to changing load and solar conditions, the voltage variability index at the inverter terminals were observed.
- Time above/below ANSI: The time duration in which any point/node on the primary feeder was below 95% of the nominal voltages.
- Losses: The impact of the smart inverters on over all feeder losses.

For volt-var smart inverter functions, a total of 187 unique control settings were analyzed. Volt-watt functions had 60 unique control settings while power factor functions had 10 control settings. Fig 6 to Fig 9 shows a summary of the improvements provided by the smart inverters based on various “feeder objectives” with the smart inverters connected to the secondary system.

	Weighting	BaseCase	PV	Voltvar	Voltwatt	PF
MeanPCCv (pu)	0.00	1.0090	1.0100	1.0137	1.0100	1.0110
Voltage Variability Index	0.00	1.0000	0.9998	1.0019	1.0000	1.0000
Tap Operations	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
Consumption (kwh)	0.00	229820.00	233620.00	234790.00	232470.00	234740.00
Losses (kwh)	0.00	12643.00	9805.90	10868.00	10699.00	10825.00
TimeAbove (secs)	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
TimeBelow (secs)	0.00	44273.00	44273.00	0.00	44273.00	44273.00
MaxV (pu)	0.00	1.0170	1.0170	1.0270	1.0170	1.0170
MinV (pu)	0.00	0.9330	0.9410	0.9510	0.9390	0.9419
Vdiff	100.00	0.0840	0.0760	0.0760	0.0780	0.0751

Fig 6: Voltage difference as key objective

	Weighting	BaseCase	PV	Voltvar	Voltwatt	PF
MeanPCCv (pu)	0.00	1.0090	1.0100	1.0137	1.0100	1.0110
Voltage Variability Index	0.00	1.0000	0.9998	1.0019	1.0000	1.0001
Tap Operations	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
Consumption (kwh)	0.00	229820.00	233620.00	234790.00	232470.00	233620.00
Losses (kwh)	0.00	12643.00	9805.90	10868.00	10699.00	9805.90
TimeAbove (secs)	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
TimeBelow (secs)	100.00	44273.00	44273.00	0.00	44273.00	44273.00
MaxV (pu)	0.00	1.0170	1.0170	1.0270	1.0170	1.0170
MinV (pu)	0.00	0.9330	0.9410	0.9510	0.9390	0.9410
Vdiff	0.00	0.0840	0.0760	0.0760	0.0780	0.0760

Fig 7: Time below ANSI as key objective

	Weighting	BaseCase	PV	Voltvar	Voltwatt	PF
MeanPCCv (pu)	0.00	1.0090	1.0100	1.0137	1.0100	1.0110
Voltage Variability Index	0.00	1.0000	0.9998	1.0070	1.0000	1.0001
Tap Operations	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
Consumption (kwh)	0.00	229820.00	233620.00	235230.00	232470.00	233620.00
Losses (kwh)	100.00	12643.00	9805.90	10857.00	10699.00	9805.90
TimeAbove (secs)	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
TimeBelow (secs)	0.00	44273.00	44273.00	0.00	44273.00	44273.00
MaxV (pu)	0.00	1.0170	1.0170	1.0270	1.0170	1.0170
MinV (pu)	0.00	0.9330	0.9410	0.9510	0.9390	0.9410
Vdiff	0.00	0.0840	0.0760	0.0760	0.0780	0.0760

Fig 8: Losses as key objective

	Weighting	BaseCase	PV	Voltvar	Voltwatt	PF
MeanPCCv (pu)	0.00	1.0090	1.0100	1.0137	1.0100	1.0110
Voltage Variability Index	100.00	1.0000	0.9998	1.0019	1.0000	1.0000
Tap Operations	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
Consumption (kwh)	0.00	229820.00	233620.00	234790.00	232470.00	234740.00
Losses (kwh)	0.00	12643.00	9805.90	10868.00	10699.00	10825.00
TimeAbove (secs)	0.00	0.0000	0.0000	0.0000	0.0000	0.0000
TimeBelow (secs)	0.00	44273.00	44273.00	0.00	44273.00	44273.00
MaxV (pu)	0.00	1.0170	1.0170	1.0270	1.0170	1.0170
MinV (pu)	0.00	0.9330	0.9410	0.9510	0.9390	0.9419
Vdiff	0.00	0.0840	0.0760	NaN	0.0780	0.0751

Fig 9: Voltage variability index as key objective

Looking at Fig 7, if the objective is to reduce the number of times in which any node across the feeder is below ANSI, volt-var control gives the best results. Volt-var control also limits the voltage variability index better than the other functions. However, fixed power factor (PF) limited the losses better than the other functions (though at the expense of increasing the time below ANSI ranges). The recommended volt-var curves and settings for the various objectives are shown in Fig 10.

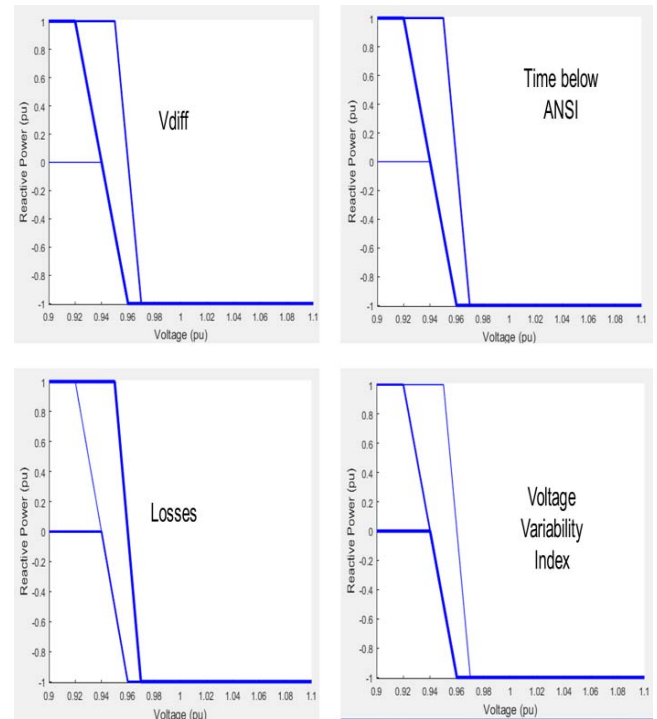


Fig 10: Volt-var smart inverter settings for various feeder objectives

The dark blue color in all the volt-var curves in Fig 10; highlights the best of the three recommended setting for the specific objective. As shown in Fig 8, the applicable settings to

reduce losses across the feeder is completely different from the recommended settings if reducing under voltage is the key objective. For Feeder XYZ, in order to reduce the losses, the smart inverters on the secondary system has to be aggressive and inductive with no deadband.

If customization of inverter settings due to expected solar/load conditions is possible, then the recommended volt-var inverter under each condition is shown in Fig 11.

There is “no specific feeder objective” considered in the curves in Fig 11, apart from the solar/load conditions. Here, the key priority is based ONLY on the solar/load conditions.

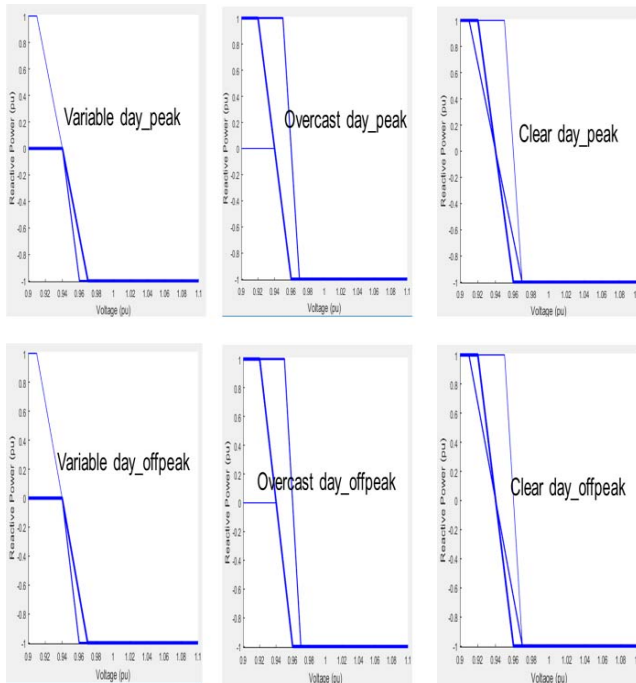


Fig 11: Volt-var smart inverter settings for various solar/load conditions

V. SUMMARY AND CONCLUSIONS

A methodology that can be applied to determine best recommended settings on secondary systems was explained. This paper discusses a quasi-static time-series simulation in order to determine optimal volt-var settings with regards to feeder metrics such as the number of regulator tap changes, losses, consumption, voltage variability, and voltage violations; for a secondary system.

The optimal control settings are compared against different system conditions to find correlations, and autonomous inverter control is shown through an annual simulation.

Generally, given the small number of inverters (relative to the total feeder load) in the study, the feeder regulating devices were generally insensitive to the slope of the volt-var curves. Increasing the number of inverters on the feeder, a change in

feeder topology, or installation of appropriate voltage regulation equipment such as regulators along with its settings may make the feeder more sensitive to setting variations. This weighting is based on voltage measurements at the medium-voltage primary. Voltage variability at individual customer service entrances may be significantly greater.

Since the specific objective for any feeder may differ, the methodology must examine a wide range of control settings.

There is a unique feeder impact for each of the different control types and settings. The best inverter settings are dependent on performance metrics (losses, voltage variability index, feeder voltage flattening, time below ANSI etc.), feeder characteristics, load and solar condition (peak load, variable solar; off peak load, clear solar etc.). The PV inverter size (kVA capacity) also plays a key role in determining the best settings.

Ongoing work is being done to improve the methodology described here to simplify the process and reduce the overall engineering time and resources needed to optimize smart inverter settings.

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